Industrial Internship Report on

"Prediction of Agriculture Crop Production in India"

Prepared by

Priyanshi Gudhka

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Prediction of Agriculture Crop Production in India". The objective of this project was to build a Machine Learning model to predict crop yield based on factors such as crop type, state, and cost parameters. A Random Forest Regressor model was used for prediction and deployed using Streamlit to provide real-time results.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

The internship program provided by Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT) and IoT Academy was a valuable learning experience. This internship focused on solving real-world problems using Machine Learning techniques.

During the six weeks of the internship, I worked on the project titled "Prediction of Agriculture Crop Production in India". The project involved understanding the problem statement, performing data analysis, preprocessing the dataset, building a machine learning model, and deploying the solution using Streamlit.

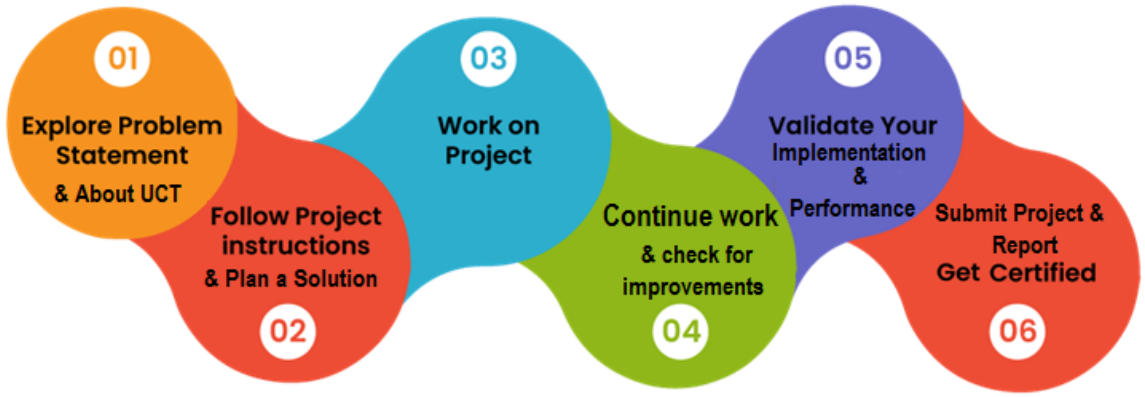
In the initial phase, I explored the dataset and performed exploratory data analysis to understand key patterns and relationships. In the next phase, I cleaned and preprocessed the data to make it suitable for model building. I then implemented a Random Forest Regressor model to predict crop yield based on various factors such as crop type, state, and cost parameters.

Further, I deployed the model using Streamlit to create an interactive web application that allows users to input data and get real-time predictions. This helped me understand the end-to-end lifecycle of a machine learning project.

This internship provided me with hands-on experience in Python, data analysis, machine learning, and deployment. It enhanced my problem-solving skills and gave me confidence in working on real-world projects.

I would like to thank Upskill Campus, UniConverge Technologies Pvt Ltd, and IoT Academy for providing this opportunity. I am also grateful to all mentors and guides who supported me throughout the internship.

I would recommend such internship programs to students as they provide practical exposure and help in building industry-relevant skills effectively.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

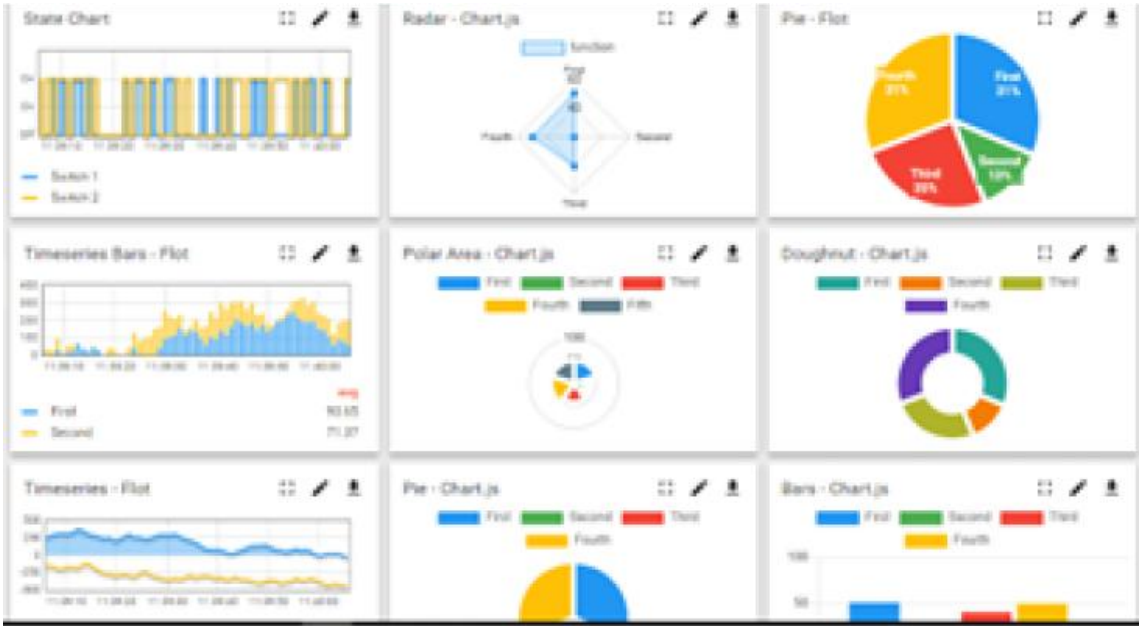
i. UCT IoT Platform (UCT Insight)

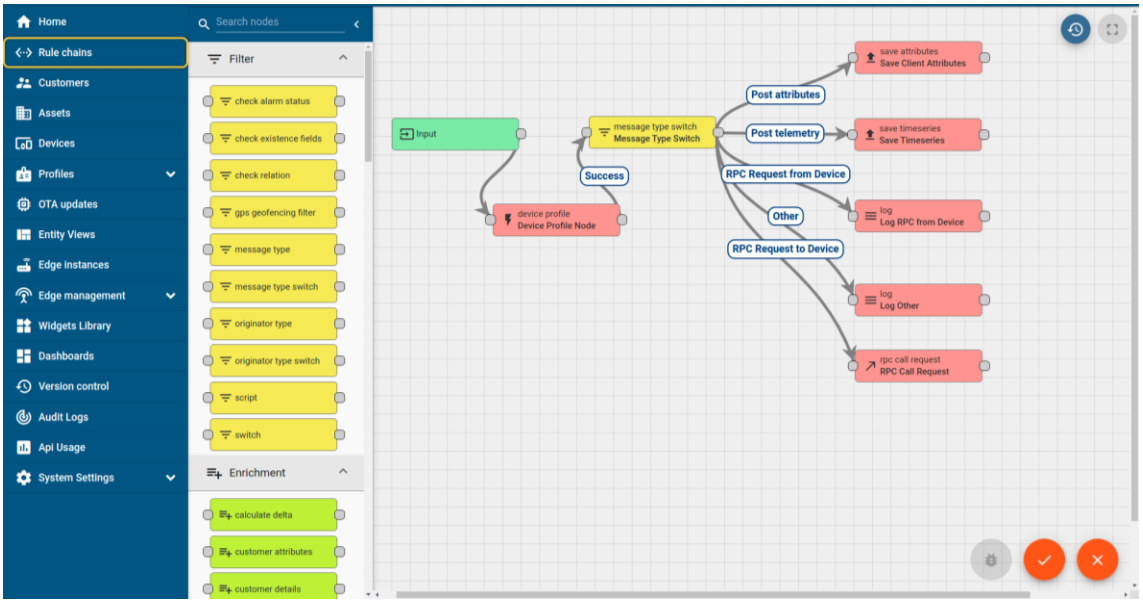
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party applications (Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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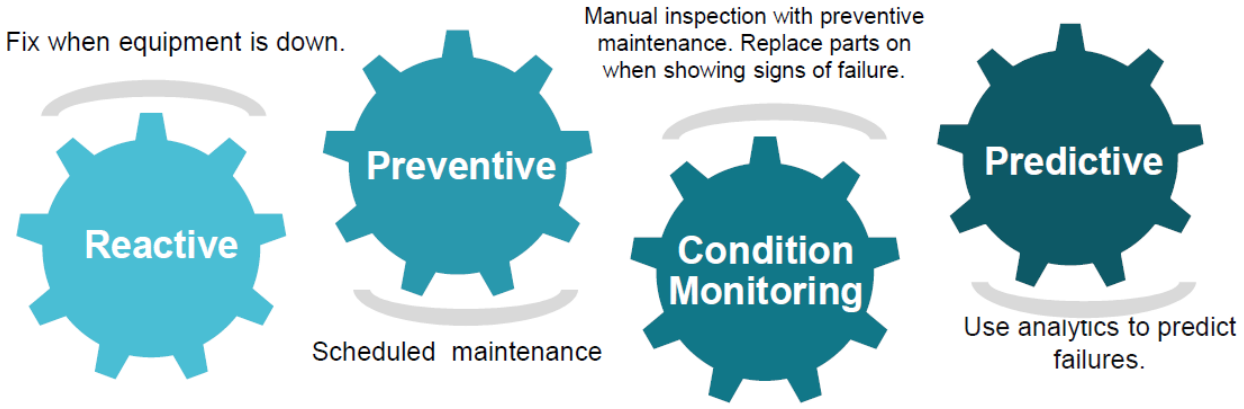
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

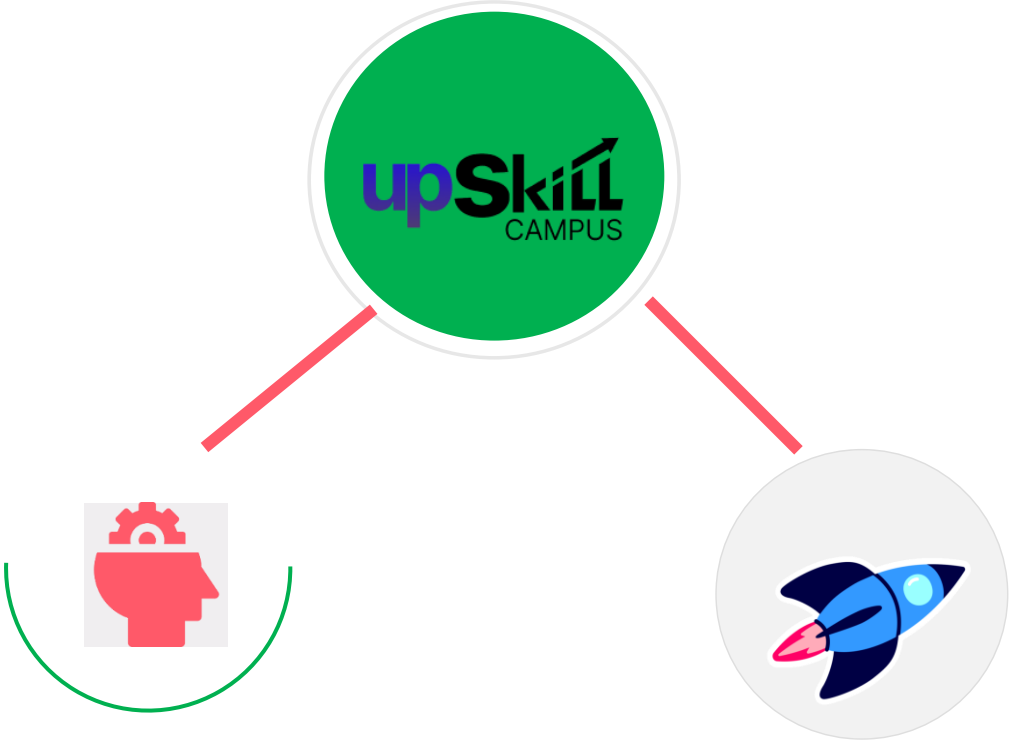
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

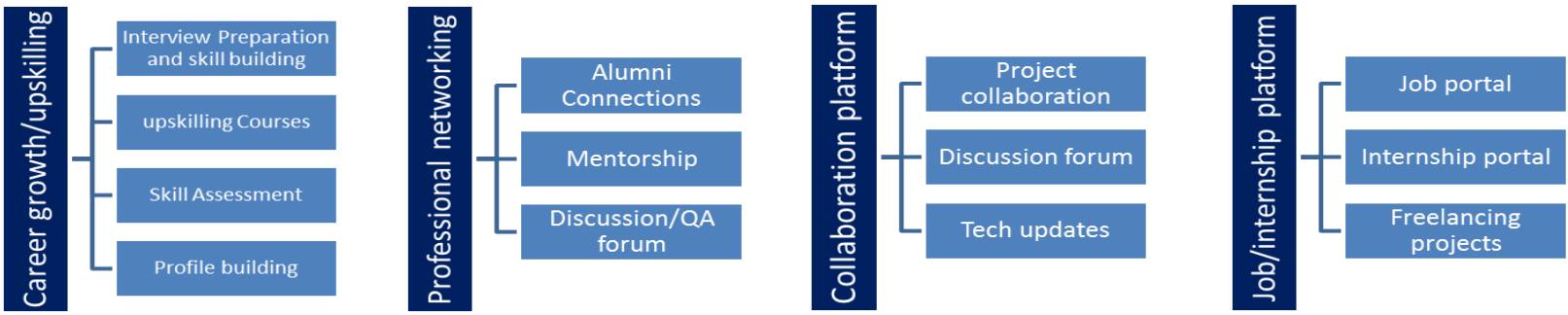
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 years

https://www.upskillcampus.com/



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Glossary

Terms	Acronym
Machine Learning	**ML**
Random Forest	**RF**
Internet of Things	**IoT**
Quintal per Hectare	Q/Ha
Cost of Cultivation	**CoC**
Streamlit	Web App Framework

3 Problem Statement

In the assigned problem statement, the objective is to predict agriculture crop production in India using Machine Learning techniques.

Agriculture plays a vital role in the Indian economy, but predicting crop yield is a complex task due to the influence of multiple factors such as crop type, geographical location, and cost of cultivation. Traditional methods of prediction are often inaccurate and do not utilize historical data effectively.

There is a need for a data-driven solution that can analyze past agricultural data and provide accurate predictions for crop yield. Such predictions can help farmers, policymakers, and stakeholders make informed decisions regarding crop planning, resource allocation, and cost management.

The aim of this project is to develop a Machine Learning model that predicts crop yield based on key parameters such as crop, state, and cost factors. The system also provides an interactive interface where users can input values and obtain real-time predictions.

4 Existing and Proposed solution

Existing Solution:

Traditional methods of predicting crop yield are mostly manual or based on basic statistical analysis. Farmers often rely on past experience or general estimates, which may not consider all influencing factors such as cost variations, regional differences, and crop-specific patterns. These methods lack accuracy and are not data-driven, making them unreliable for decision-making.

Limitations of Existing System:

- Does not use Machine Learning techniques
- Low prediction accuracy
- No real-time prediction capability
- Does not consider multiple influencing factors together
- No user-friendly interface for general users

Proposed Solution:

The proposed system uses Machine Learning techniques to predict crop yield based on various input parameters such as crop type, state, and cost factors. A Random Forest Regressor model is used to improve prediction accuracy.

The system also includes a user-friendly web application built using Streamlit, where users can input data and get instant predictions.

Value Addition:

- Provides accurate predictions using Machine Learning
- Easy-to-use interactive interface for users
- Helps farmers and stakeholders in better decision-making
- Reduces dependency on traditional estimation methods
- Can be extended with more datasets for improved performance

4.1 Code submission (GitHub Link)

https://github.com/priyanshigudhka-gif/upskillcampus

4.2 Report submission (GitHub Link)

https://raw.githubusercontent.com/priyanshigudhka-gif/upskillcampus/main/AgricultureCropYieldPredictionSystem_Priyanshi%20Gudhka_USC_UCT.pdf

5 Proposed Design/ Model

The proposed system is designed to predict crop yield using Machine Learning techniques. The system follows a structured pipeline starting from data collection to deployment.

The workflow includes data preprocessing, feature selection, model training, and prediction using a trained model. A Random Forest Regressor algorithm is used to improve prediction accuracy and handle complex relationships between features.

5.1 High Level Diagram (if applicable)

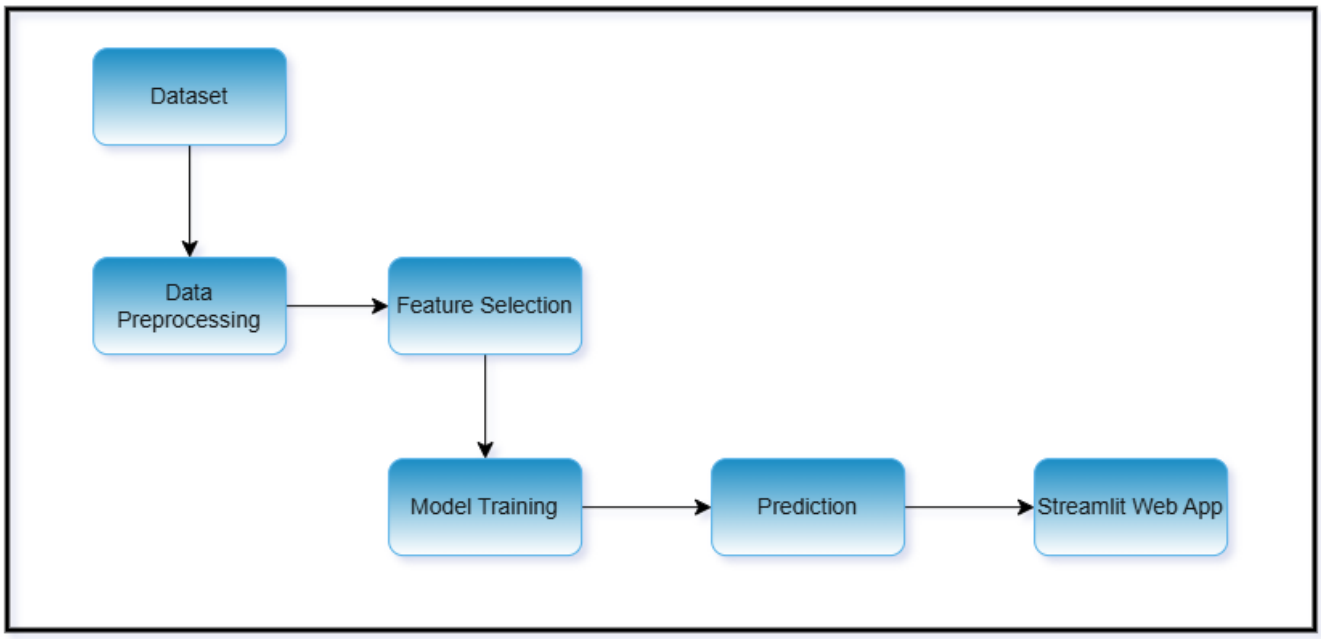


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

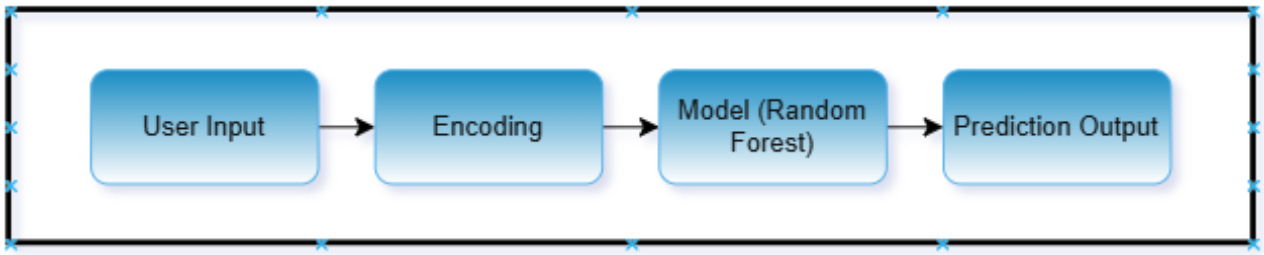


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

The system provides a web-based interface using Streamlit where users can input crop details such as crop type, state, and cost parameters. Based on the input, the system predicts crop yield using a trained machine learning model.

Below is the interface of the developed system.

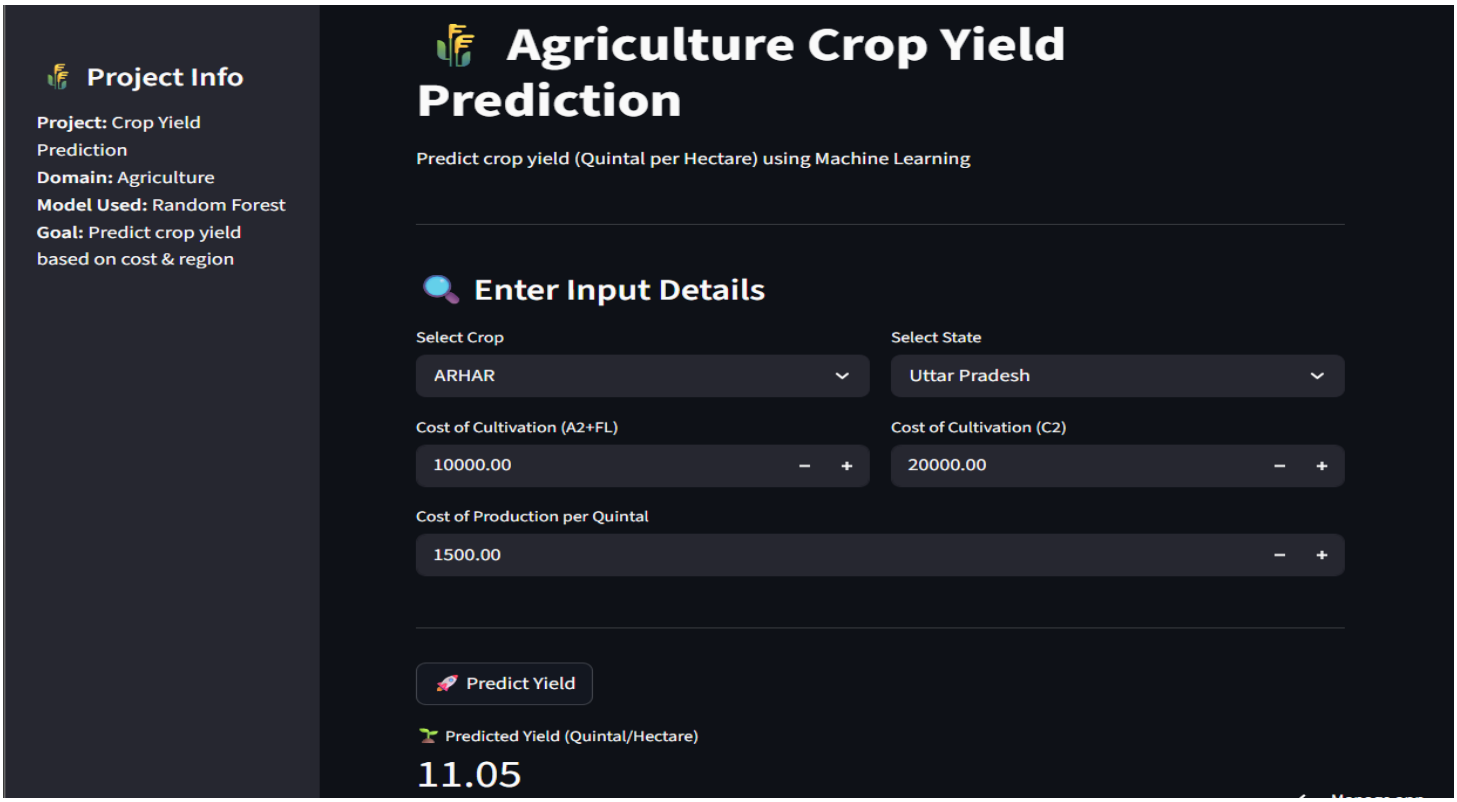


Figure 3: Input Interface of Crop Yield Prediction System

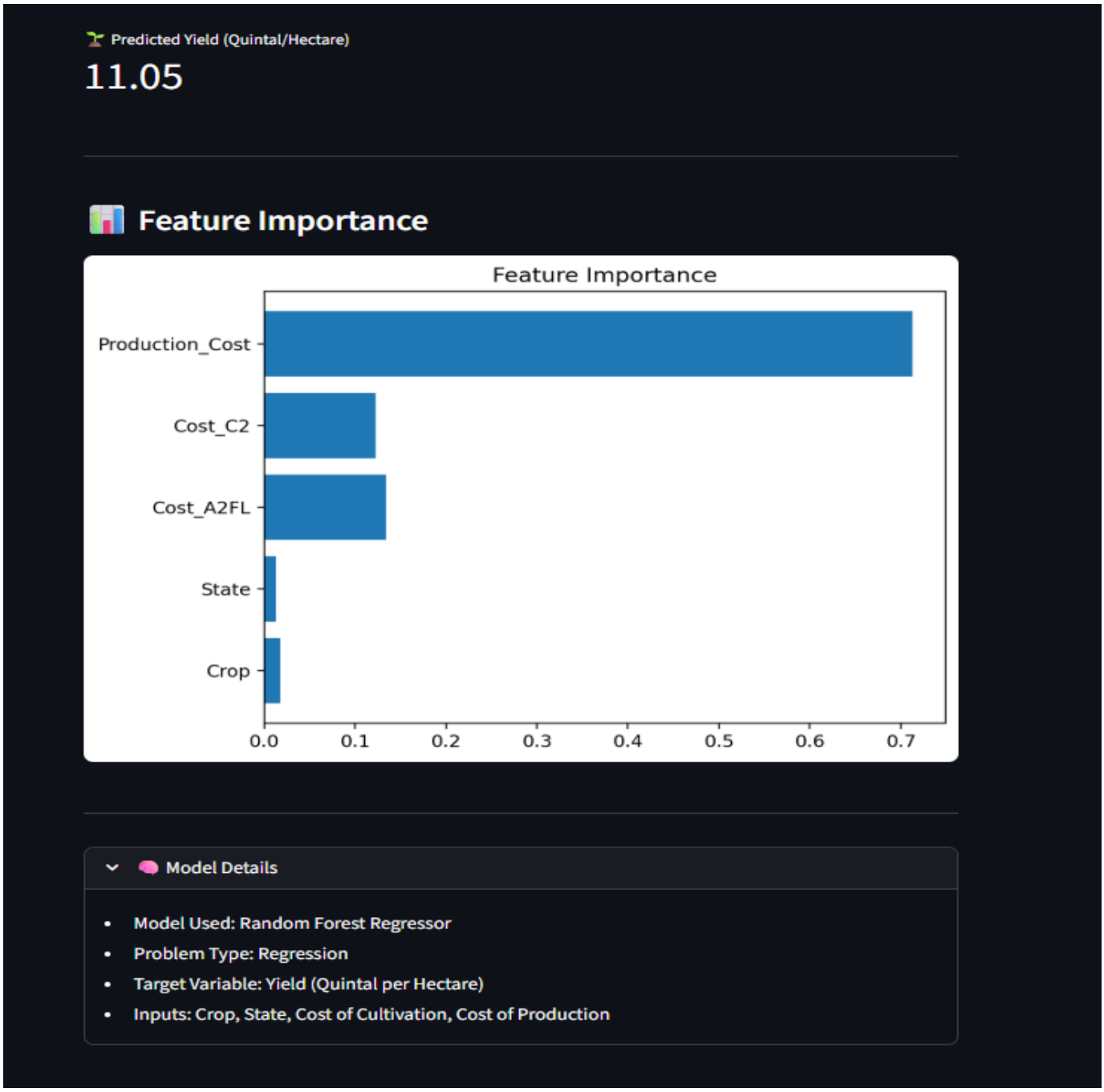


Figure 4: Prediction Output and Feature Importance

6 Performance Test

The performance of the developed system was evaluated based on usability, response time, and prediction accuracy.

This section evaluates the performance of the developed machine learning model and the deployed web application.

The system was tested based on various constraints such as accuracy, response time, and usability. The model is designed to handle real-world agricultural data efficiently and provide quick predictions to users.

The performance of the system is analyzed using different test cases and evaluation metrics to ensure reliability and effectiveness.

6.1 Test Plan/ Test Cases

Different test cases were considered by providing various combinations of input values such as crop type, state, cost of cultivation, and cost of production.

The system was tested with:

- Different crop types (e.g., Rice, Wheat, Cotton)
- Different states across India
- Varying cost values to check prediction stability

These test cases ensured that the model performs consistently across different scenarios.

6.2 Test Procedure

The testing procedure involved entering input values into the Streamlit web application interface.

Steps followed:

1. Select crop and state from dropdown
2. Enter cost parameters
3. Click on "Predict Yield"
4. Observe predicted output

The system processes the input through the trained Random Forest model and displays the predicted yield instantly.

6.3 Performance Outcome

The model showed good prediction capability based on the available dataset.

- Predictions were generated instantly (low response time)
- Model handled different inputs effectively
- Feature importance graph helped understand key influencing factors

Overall, the system performs efficiently and provides reliable predictions for agricultural crop yield.

7 My learnings

During this internship, I gained valuable hands-on experience in the field of Machine Learning and Data Analysis. I learned how to work with real-world datasets, perform data preprocessing, and apply various machine learning algorithms to solve practical problems.

I developed a complete end-to-end project, starting from data understanding and cleaning to model building using the Random Forest Regressor and finally deploying the solution using Streamlit. This helped me understand the full lifecycle of a machine learning project.

I also improved my technical skills in Python, including libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. Additionally, I learned how to visualize data and interpret results effectively.

Apart from technical skills, this internship enhanced my problem-solving ability, logical thinking, and project development skills. I also gained experience in using GitHub for version control and project management.

Overall, this internship has significantly contributed to my career growth and has strengthened my interest in the field of Data Science and Machine Learning.

Key Learnings:

- End-to-end ML project development
- Data preprocessing and feature engineering
- Model deployment using Streamlit
- GitHub project management

8 Future work scope

The current system successfully predicts agriculture crop yield using machine learning techniques. However, there are several improvements and extensions that can be implemented in the future.

The model can be enhanced by incorporating more features such as weather data (temperature, rainfall, humidity), soil quality parameters, and fertilizer usage, which can improve prediction accuracy significantly.

Advanced machine learning techniques such as Deep Learning models (ANN, LSTM) can also be explored for better performance and handling complex patterns in agricultural data.

The application can be further improved by integrating real-time data sources through APIs, allowing dynamic and up-to-date predictions.

A mobile application can also be developed to make the system more accessible to farmers and stakeholders in rural areas.

Additionally, the system can be expanded to provide crop recommendation, price prediction, and decision support systems for better agricultural planning.

These enhancements can make the system more robust, scalable, and impactful in real-world agricultural scenarios.

These improvements can further enhance the scalability and real-world impact of the system.